

Solving ODE Systems

*Homogeneous Constant-Coefficient Linear Systems:
Distinct Real, Repeated, & Complex Eigenvalues with Phase Portraits*

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$\mathbf{x}' = A\mathbf{x}$ • Distinct Real Eigenvalues • Nodes & Saddles • Repeated Eigenvalues • Improper Nodes • Complex Eigenvalues • Spirals & Centers • Trace–Determinant Plane • Phase Portraits

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1 Part I | Condensed Cheat Sheet

1.1 The Setup: $\mathbf{x}' = A\mathbf{x}$ and the Phase Plane

A homogeneous constant-coefficient system $\mathbf{x}' = A\mathbf{x}$ is solved by feeding the eigenpairs of A into exponential solutions $\boldsymbol{\xi} e^{\lambda t}$. What changes from problem to problem is only the *type* of eigenvalues — and that type also fixes the geometry of the trajectories in the phase plane. When $\det A \neq 0$ the origin is the only equilibrium, and the eigenvalues classify it completely.

Master strategy. Seek $\mathbf{x} = \boldsymbol{\xi} e^{\lambda t}$; this forces the eigenvalue problem $(A - \lambda I)\boldsymbol{\xi} = \mathbf{0}$. Build n independent solutions, then

$$\mathbf{x}(t) = c_1 \mathbf{x}^{(1)}(t) + \cdots + c_n \mathbf{x}^{(n)}(t).$$

Three cases for a 2×2 matrix: **distinct real**, **repeated real**, **complex** eigenvalues.

1.2 Case 1 — Distinct Real Eigenvalues (Nodes & Saddles)

With $\lambda_1 \neq \lambda_2$ both real, the two eigenvectors are independent and the general solution is a clean superposition. The eigenvector directions are themselves straight-line trajectories.

General solution ($\lambda_1 \neq \lambda_2$ real):

$$\mathbf{x}(t) = c_1 \boldsymbol{\xi}^{(1)} e^{\lambda_1 t} + c_2 \boldsymbol{\xi}^{(2)} e^{\lambda_2 t}.$$

Phase portrait.

- Both negative $\Rightarrow$ *nodal sink* (stable node) — curves enter the origin tangent to the *slow* direction (smaller $|\lambda|$).
- Both positive $\Rightarrow$ *nodal source* (unstable node) — same picture, arrows reversed.
- Opposite signs $\Rightarrow$ *saddle* (unstable) — eigendirections are the separatrices.

Worked Example — a stable node. Solve $\mathbf{x}' = \begin{pmatrix} -3 & 1 \\ 1 & -3 \end{pmatrix} \mathbf{x}$.

Step 1. Eigenvalues ($\text{tr } A = -6$, $\det A = 8$):

$$\lambda^2 + 6\lambda + 8 = (\lambda + 2)(\lambda + 4) = 0 \implies \lambda = -2, -4.$$

Step 2. Eigenvectors — $\lambda = -2$: $\boldsymbol{\xi}^{(1)} = (1, 1)^\top$; $\lambda = -4$: $\boldsymbol{\xi}^{(2)} = (1, -1)^\top$.

Step 3. General solution:

$$\mathbf{x}(t) = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{-2t} + c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-4t}.$$

Step 4. Both eigenvalues negative — a *stable node*; trajectories approach the origin tangent to the slow direction $(1, 1)^\top$ (eigenvalue -2).

1.3 Case 2 — Repeated Real Eigenvalue (Proper & Improper Nodes)

A double root λ splits into two sub-cases according to how many independent eigenvectors it has.

Complete (two independent eigenvectors — forces $A = \lambda I$): every vector is an eigenvector, so

$$\mathbf{x}(t) = e^{\lambda t} \mathbf{c} \quad (\text{any constant vector } \mathbf{c}) \Rightarrow \text{proper / star node.}$$

Defective (only one eigenvector $\boldsymbol{\xi}$): find a *generalized eigenvector* $\boldsymbol{\eta}$ from $(A - \lambda I)\boldsymbol{\eta} = \boldsymbol{\xi}$, then

$$\mathbf{x}(t) = c_1 \boldsymbol{\xi} e^{\lambda t} + c_2 (\boldsymbol{\xi} t + \boldsymbol{\eta}) e^{\lambda t} \Rightarrow \text{improper node.}$$

Stable if $\lambda < 0$, unstable if $\lambda > 0$; every trajectory becomes tangent to $\pm \boldsymbol{\xi}$.

Worked Example — a defective (improper) node. Solve $\mathbf{x}' = \begin{pmatrix} -1 & -1 \\ 1 & -3 \end{pmatrix} \mathbf{x}$.

Step 1. Eigenvalue ($\text{tr } A = -4$, $\det A = 4$):

$$\lambda^2 + 4\lambda + 4 = (\lambda + 2)^2 = 0 \Rightarrow \lambda = -2 \text{ (double).}$$

Step 2. One eigenvector — $(A + 2I)\boldsymbol{\xi} = \mathbf{0} \Rightarrow \xi_1 - \xi_2 = 0$:

$$\boldsymbol{\xi} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Step 3. Generalized eigenvector — $(A + 2I)\boldsymbol{\eta} = \boldsymbol{\xi} \Rightarrow \eta_1 - \eta_2 = 1$:

$$\boldsymbol{\eta} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Step 4. General solution — a *stable improper node*:

$$\mathbf{x}(t) = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{-2t} + c_2 \left[\begin{pmatrix} 1 \\ 1 \end{pmatrix} t + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right] e^{-2t}.$$

1.4 Case 3 — Complex Eigenvalues (Spirals & Centers)

A complex pair $\lambda = \alpha \pm i\beta$ produces oscillation. Take one complex solution $\boldsymbol{\xi} e^{\lambda t}$ and split it into real and imaginary parts to obtain two real solutions.

General solution ($\lambda = \alpha \pm i\beta$, $\boldsymbol{\xi} = \mathbf{a} + i\mathbf{b}$):

$$\mathbf{x}(t) = c_1 e^{\alpha t} (\mathbf{a} \cos \beta t - \mathbf{b} \sin \beta t) + c_2 e^{\alpha t} (\mathbf{a} \sin \beta t + \mathbf{b} \cos \beta t).$$

Phase portrait. $\alpha < 0 \Rightarrow \text{stable spiral}$; $\alpha > 0 \Rightarrow \text{unstable spiral}$; $\alpha = 0 \Rightarrow \text{center}$ (closed orbits, neutrally stable). Rotation sense comes from the field at one test point.

Worked Example — a stable spiral. Solve $\mathbf{x}' = \begin{pmatrix} -1 & -2 \\ 2 & -1 \end{pmatrix} \mathbf{x}$.

Step 1. Eigenvalues ($\text{tr } A = -2$, $\det A = 5$):

$$\lambda^2 + 2\lambda + 5 = 0 \implies \lambda = -1 \pm 2i, \quad \alpha = -1, \beta = 2.$$

Step 2. Eigenvector for $\lambda = -1 + 2i$ (row $\xi_1 - 2i\xi_2 = 0 \dots$ here $-2i\xi_1 - 2\xi_2 = 0$):

$$\boldsymbol{\xi} = \begin{pmatrix} 1 \\ -i \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + i \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \quad \mathbf{a} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}.$$

Step 3. General solution — a *stable spiral* ($\alpha = -1 < 0$):

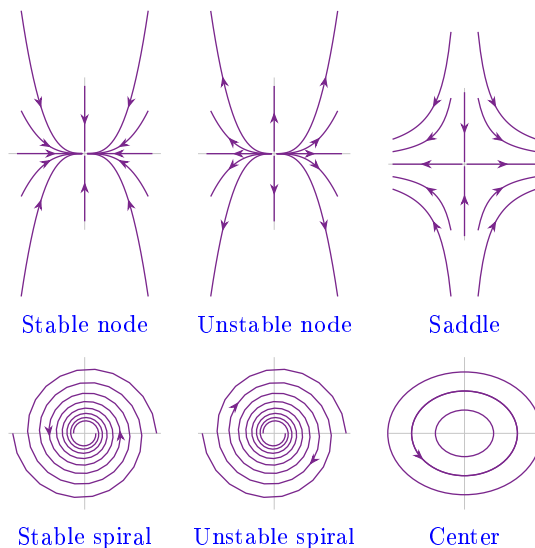
$$\mathbf{x}(t) = e^{-t} \left[c_1 \begin{pmatrix} \cos 2t \\ \sin 2t \end{pmatrix} + c_2 \begin{pmatrix} \sin 2t \\ -\cos 2t \end{pmatrix} \right].$$

1.5 The Trace–Determinant Plane

Every 2×2 classification can be read from just two numbers — $\tau = \text{tr } A$ and $\Delta = \det A$ — without computing eigenvectors.

With $\lambda = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2}$:

- $\Delta < 0 \implies$ **saddle** (always unstable).
- $\Delta > 0$, $\tau^2 - 4\Delta > 0 \implies$ **node** ($\tau < 0$ stable, $\tau > 0$ unstable).
- $\Delta > 0$, $\tau^2 - 4\Delta < 0 \implies$ **spiral** ($\tau < 0$ stable, $\tau > 0$ unstable).
- $\Delta > 0$, $\tau = 0 \implies$ **center**.
- $\tau^2 - 4\Delta = 0 \implies$ repeated root: proper or improper **node**.



Note. A repeated eigenvalue lands on the boundary parabola $\tau^2 = 4\Delta$ that separates nodes from spirals — so an improper node looks like a node caught just before it spirals.

2 Part II | Review Guide

For each system, find the general solution (or as requested) and classify the equilibrium at the origin. Solutions follow in Part III.

1. Solve $\mathbf{x}' = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix} \mathbf{x}$ and classify the origin.
2. Without solving, classify the origin (type and stability) for each matrix using the trace–determinant test: (a) $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, (b) $\begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$, (c) $\begin{pmatrix} -1 & -1 \\ 1 & -1 \end{pmatrix}$.
3. Solve $\mathbf{x}' = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \mathbf{x}$ and classify the origin.
4. Solve $\mathbf{x}' = \begin{pmatrix} 3 & -2 \\ -2 & 3 \end{pmatrix} \mathbf{x}$ and classify the origin.
5. Solve $\mathbf{x}' = \begin{pmatrix} -2 & -3 \\ 3 & -2 \end{pmatrix} \mathbf{x}$ and classify the origin.
6. Solve $\mathbf{x}' = \begin{pmatrix} 0 & 4 \\ -1 & 0 \end{pmatrix} \mathbf{x}$ and classify the origin.
7. Solve $\mathbf{x}' = \begin{pmatrix} 2 & 1 \\ -1 & 4 \end{pmatrix} \mathbf{x}$ (a repeated-eigenvalue system) and classify the origin.
8. Solve the IVP $\mathbf{x}' = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix} \mathbf{x}$, $\mathbf{x}(0) = \begin{pmatrix} 0 \\ 4 \end{pmatrix}$, and classify the origin.
9. Solve the IVP $\mathbf{x}' = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \mathbf{x}$, $\mathbf{x}(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, and classify the origin.
10. Solve the IVP $\mathbf{x}' = \begin{pmatrix} 2 & 1 \\ -1 & 4 \end{pmatrix} \mathbf{x}$, $\mathbf{x}(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ (same matrix as Problem 7), and describe the trajectory.

3 Part III | Answer Key & Worked Solutions

Solution 1

$\text{tr } A = -4, \det A = 3$:

$$\lambda^2 + 4\lambda + 3 = (\lambda + 1)(\lambda + 3) = 0 \implies \lambda = -1, -3.$$

$\lambda = -1$: $\xi^{(1)} = (1, 1)^\top$; $\lambda = -3$: $\xi^{(2)} = (1, -1)^\top$.

$$\mathbf{x}(t) = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{-t} + c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-3t}.$$

Both eigenvalues negative $\Rightarrow$ **stable node**.

Solution 2

(a) $\tau = 4, \Delta = 3, \tau^2 - 4\Delta = 4 > 0 \Rightarrow$ **unstable node**.

(b) $\tau = 0, \Delta = 4 > 0 \Rightarrow$ **center**.

(c) $\tau = -2, \Delta = 2, \tau^2 - 4\Delta = -4 < 0 \Rightarrow$ **stable spiral**.

Solution 3

$\text{tr } A = 2, \det A = -3$:

$$\lambda^2 - 2\lambda - 3 = (\lambda - 3)(\lambda + 1) = 0 \implies \lambda = 3, -1.$$

$\lambda = 3$: $\xi^{(1)} = (1, 1)^\top$; $\lambda = -1$: $\xi^{(2)} = (1, -1)^\top$.

$$\mathbf{x}(t) = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{3t} + c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-t}.$$

Opposite signs $\Rightarrow$ **saddle** (unstable).

Solution 4

$\text{tr } A = 6, \det A = 5$:

$$\lambda^2 - 6\lambda + 5 = (\lambda - 1)(\lambda - 5) = 0 \implies \lambda = 1, 5.$$

$\lambda = 1$: $\xi^{(1)} = (1, 1)^\top$; $\lambda = 5$: $\xi^{(2)} = (1, -1)^\top$.

$$\mathbf{x}(t) = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t + c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{5t}.$$

Both positive $\Rightarrow$ **unstable node**.

Solution 5

$\text{tr } A = -4, \det A = 13$:

$$\lambda^2 + 4\lambda + 13 = 0 \implies \lambda = -2 \pm 3i, \quad \alpha = -2, \beta = 3.$$

Eigenvector for $\lambda = -2 + 3i$: $-3i\xi_1 - 3\xi_2 = 0 \Rightarrow \xi_2 = -i\xi_1$, so $\xi = (1, -i)^\top$ with $\mathbf{a} = (1, 0)^\top$, $\mathbf{b} = (0, -1)^\top$.

$$\mathbf{x}(t) = e^{-2t} \left[c_1 \begin{pmatrix} \cos 3t \\ \sin 3t \end{pmatrix} + c_2 \begin{pmatrix} \sin 3t \\ -\cos 3t \end{pmatrix} \right].$$

$\alpha = -2 < 0 \Rightarrow$ **stable spiral**.

Solution 6

$\text{tr } A = 0, \det A = 4$:

$$\lambda^2 + 4 = 0 \implies \lambda = \pm 2i, \quad \alpha = 0, \beta = 2.$$

Eigenvector for $\lambda = 2i$ (row $-\xi_1 - 2i\xi_2 = 0$): $\boldsymbol{\xi} = (-2i, 1)^\top$, so $\mathbf{a} = (0, 1)^\top$, $\mathbf{b} = (-2, 0)^\top$.

$$\mathbf{x}(t) = c_1 \begin{pmatrix} 2 \sin 2t \\ \cos 2t \end{pmatrix} + c_2 \begin{pmatrix} -2 \cos 2t \\ \sin 2t \end{pmatrix}.$$

$\alpha = 0 \Rightarrow$ **center** (closed elliptical orbits).

Solution 7

$\text{tr } A = 6, \det A = 9$:

$$\lambda^2 - 6\lambda + 9 = (\lambda - 3)^2 = 0 \implies \lambda = 3 \text{ (double)}.$$

One eigenvector — $(A - 3I)\boldsymbol{\xi} = \mathbf{0} \Rightarrow -\xi_1 + \xi_2 = 0$:

$$\boldsymbol{\xi} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Generalized eigenvector — $(A - 3I)\boldsymbol{\eta} = \boldsymbol{\xi} \Rightarrow -\eta_1 + \eta_2 = 1$:

$$\boldsymbol{\eta} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

$$\mathbf{x}(t) = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{3t} + c_2 \left[\begin{pmatrix} 1 \\ 1 \end{pmatrix} t + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right] e^{3t}.$$

$\lambda = 3 > 0$, one eigenvector $\Rightarrow$ **unstable improper node**.

Solution 8

$\text{tr } A = 2, \det A = -3$: $\lambda^2 - 2\lambda - 3 = (\lambda - 3)(\lambda + 1) = 0 \Rightarrow \lambda = 3, -1$, with $\boldsymbol{\xi}^{(1)} = (1, 2)^\top$, $\boldsymbol{\xi}^{(2)} = (1, -2)^\top$.

$$\mathbf{x}(t) = c_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{3t} + c_2 \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{-t}.$$

Initial condition — $c_1 + c_2 = 0, 2c_1 - 2c_2 = 4 \Rightarrow c_1 = 1, c_2 = -1$:

$$\mathbf{x}(t) = \begin{pmatrix} e^{3t} - e^{-t} \\ 2e^{3t} + 2e^{-t} \end{pmatrix}.$$

Opposite signs $\Rightarrow$ **saddle**.

Solution 9

$\text{tr } A = 2, \det A = 2$:

$$\lambda^2 - 2\lambda + 2 = 0 \implies \lambda = 1 \pm i, \quad \alpha = 1, \beta = 1.$$

Eigenvector for $\lambda = 1 + i$ (row $\xi_1 - i\xi_2 = 0$): $\boldsymbol{\xi} = (i, 1)^\top$, so $\mathbf{a} = (0, 1)^\top$, $\mathbf{b} = (1, 0)^\top$.

$$\mathbf{x}(t) = e^t \left[c_1 \begin{pmatrix} -\sin t \\ \cos t \end{pmatrix} + c_2 \begin{pmatrix} \cos t \\ \sin t \end{pmatrix} \right].$$

Initial condition $\mathbf{x}(0) = (1, 0)^\top$ gives $c_2 = 1$, $c_1 = 0$:

$$\mathbf{x}(t) = e^t \begin{pmatrix} \cos t \\ \sin t \end{pmatrix}.$$

$\alpha = 1 > 0 \Rightarrow$ **unstable spiral**.

Solution 10

Using the general solution from Solution 7, apply $\mathbf{x}(0) = (1, 0)^\top$:

$$c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \implies c_1 = 1, \ c_2 = -1.$$

$$\mathbf{x}(t) = \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{3t} - \left[\begin{pmatrix} 1 \\ 1 \end{pmatrix} t + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right] e^{3t} = e^{3t} \begin{pmatrix} 1-t \\ -t \end{pmatrix}.$$

The trajectory leaves the origin (since $\lambda = 3 > 0$); as $t \rightarrow \infty$ the direction $(1-t, -t)$ aligns with $(-1, -1) \parallel \boldsymbol{\xi}$ — it leaves tangent to the single eigendirection, the signature of an **unstable improper node**.